

RayGlides Cooling Controller - Complete Bill of Materials (BOM)

Ref	Component	Value	Manufacturer	Part Number	Package	Quantity	Purpose
F1	Input Fuse	2A Slow-Blow	Littelfuse	0452002.MRL	UMF (SMD 2410	1	48V overcurrent protection
D1	Reverse Polarity Diode	100V 1A Schottky	Diodes Inc.	DFLS1100-7	PowerDI-123	1	Input reverse connection protection
D2	TVS Diode	SMCJ58A (Uni-dir)	Littelfuse	SMCJ58A	SMC (DO-214AE	1	Vehicle supply surge clamping (clamping at 93V)
C1	Input Filter Cap 1	10uF 100V	Panasonic	EEU-FR2A100	Radial (5x11mm)	1	Pi filter high-voltage bulk capacitor
C2	Input Filter Cap 2	10uF 100V	Panasonic	EEU-FR2A100	Radial (5x11mm)	1	Pi filter output bulk capacitor
L1	Filter Inductor	47uH	Bourns	SRR1260-470M	SMD 12x12mm	1	Pi filter ripple suppression inductor
U1	12V Buck Regulator	LM5017 Buck Converter	Texas Instruments	LM5017MR/NOPB	SO PowerPAD-8	1	Nominal 12V intermediate step-down supply
L2	Power Inductor	220uH	Bourns	SRR1260-221K	SMD 12x12mm	1	LM5017 output ripple suppression inductor
C3	Buck Bypass Cap	1uF 100V	Murata	GRM31CR72A105KA01L	1206 (SMD)	1	LM5017 high-frequency bypass capacitor
C4	Boot Cap	0.01uF 50V	Yageo	CC0603KRX7R9BB103	0603 (SMD)	1	LM5017 floating gate-drive boot capacitor
C5	Buck Output Cap	10uF 25V	Murata	GRM21BR71E106KA73L	0805 (SMD)	1	12V intermediate rail filter capacitor
R1	Feedback Resistor High	88.7k 1%	Yageo	RC0603FR-0788K7L	0603 (SMD)	1	LM5017 output voltage setting (12V feedback loop)
R2	Feedback Resistor Low	10k 1%	Yageo	RC0603FR-0710KL	0603 (SMD)	1	LM5017 feedback divider lower resistor
D3	Boot Diode	Fast Switching	ON Semiconductor	1N4148W-7-F	SOD-123	1	LM5017 boot capacitor charging diode
U2	5V Buck Regulator	AP63205 2A Buck	Diodes Inc.	AP63205WU-7	TSOT26	1	Secondary high-efficiency step-down supply
U3	3.3V LDO	AP2112K-3.3 (600mA)	Diodes Inc.	AP2112K-3.3TRG1	SOT-25	1	Low-noise MCU logic rail LDO regulator
Q1	Switching MOSFET	N-Ch 30V 5.7A	Alpha & Omega	AO3400A	SOT-23	1	Low-side fan PWM control switch
R4	Gate Resistor	100R 5%	Yageo	RC0603JR-07100RL	0603 (SMD)	1	MOSFET switching current limit resistor
R5	Gate Pull-down	10k 5%	Yageo	RC0603JR-0710KL	0603 (SMD)	1	MOSFET default-OFF gate state pull-down
D4	Flyback Diode	1N4148 Fast	ON Semiconductor	1N4148W-7-F	SOD-123	1	Fan motor inductive kickback clamping
U4	Temp Sensor	LM35 Active Linear	Texas Instruments	LM35DT/NOPB	TO-220-3	1	Heatsink temperature analog active sensor
R6	ADC Resistor	1k 1%	Yageo	RC0603FR-071KL	0603 (SMD)	1	ADC input filter series resistor
R7	Sensor Load Resistor	200R 1%	Yageo	RC0603FR-07200RL	0603 (SMD)	1	LM35 output load pull-down resistor

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C9	ADC Filter Cap	100nF 50V	Yageo	CC0603KRX7R9BB104	0603 (SMD)	1	ADC analog high-frequency noise filter capacitor
D5	ADC TVS Clamp	3.3V Zener	ON Semiconductor	BZX84C3V3LT1G	SOT-23	1	MCU ADC pin input voltage protection clamp
C7	MCU Decoupling Cap	100nF 50V	Yageo	CC0603KRX7R9BB104	0603 (SMD)	1	ESP32-S3 VDD high-frequency decoupling capacitor
C8	Sensor Decoupling Cap	100nF 50V	Yageo	CC0603KRX7R9BB104	0603 (SMD)	1	LM35 VCC high-frequency decoupling capacitor
J1	Vehicle Power Conn	Molex Micro-Fit 3.0	Molex	43650-0215	SMD Vertical	1	Primary vehicle power harness socket
J2	Fan Output Conn	Molex Micro-Fit 3.0	Molex	43650-0215	SMD Vertical	1	Cooling fan wiring harness socket
J3	Temp Sensor Conn	Molex Micro-Fit 3.0	Molex	43650-0315	SMD Vertical	1	Sensor extension cable harness socket
U5	ADC Reference	1.0V 0.1% Shunt	Texas Instruments	LM4040AIM3-1.0/NOPB	SOT-23	1	MCU ADC precision analog reference diode
U6	CAN Transceiver	High Speed Transceiver	NXP	TJA1050T/3	SOIC-8	1	Automotive CAN bus physical layer transceiver
ESD1	USB ESD Protection	Low Capacitance Clamp	STMicroelectronics	USBLC6-2SC6	SOT-23-6	1	Native USB D+/D- lines ESD protection
D6	Backpower Diode	20V 2A Schottky	ON Semiconductor	MBRA210LT3G	SMA	1	USB-C programming VBUS back-power protection
J5	USB Interface	USB Type-C Connector	Amphenol	10129424-00021LF	16-pin SMT	1	ESP32-S3 direct USB-C JTAG/serial flashing port
R8 R9	CC Pull-downs	5.1k 5%	Yageo	RC0603JR-075K1L	0603	2	USB-C CC line pull-down configuration resistors
R10 R11	USB Damping	22R 1%	Yageo	RC0603FR-0722RL	0603	2	High-speed native USB signal damping resistors
R12 R13	CAN Termination	60R 1%	Yageo	RC0603FR-0760RL	0603	2	Split CAN bus 120R termination resistors
C11	VBUS Filter Cap	4.7uF 10V	Murata	GRM188R61A475KE15D	0603	1	USB-C power input filter capacitor
C12	CAN Filter Cap	4.7nF 50V	Yageo	CC0603KRX7R9BB472	0603	1	Split CAN termination common-mode filter capacitor
J4	CAN Bus Output	Molex Micro-Fit 3.0	Molex	43650-0215	SMD Vertical	1	CAN interface vehicle bus wire connector